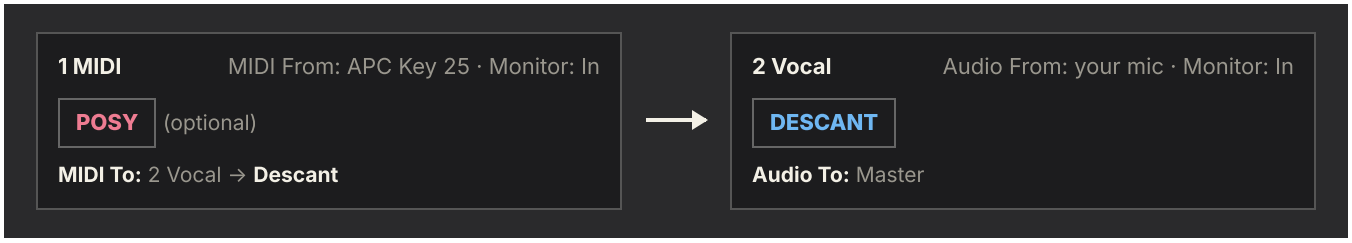
DESCANT

Getting started: a vocoder, and a choir of your own voice, in Ableton Live



Your voice comes in on audio, the notes come in on MIDI

Descant is an effect for your microphone. Sing into it and play notes on a keyboard, or let Posy send it chords: the **vocoder** puts your words on a synth, and four **harmony** voices and a **double** sing along in your own voice. It can also follow your singing with no keyboard at all. This guide gets it working in Live in a few minutes, then shows you round.



Two tracks: one for your voice, one for the notes. Descant sits on the voice track.

You'll need

- Ableton Live 11 or 12 (any edition that loads VST3 plug-ins).
- Descant built and installed. Building it copies **Descant.vst3** into ~/Library/Audio/Plug-Ins/VST3 for you; a release download goes in the same folder.
- A microphone into your audio interface, and **headphones**.
- Something to play notes: a MIDI keyboard such as an APC Key 25, a MIDI clip, or Posy.

> **Headphones, not speakers.** With speakers the microphone hears the vocoder and the harmonies and sings them back in, and it can howl. Keep the speakers down while you sing.

Set up

1 Let Live see Descant

Open **Live → Settings** (Cmd+comma) and choose the **Plug-Ins** tab. Turn on **Use VST3 Plug-In System Folders**, then press **Rescan**. In the browser, open **Plug-Ins**. Descant is under **Taddei**.

2 A track for your voice

Make an audio track. Set **Audio From** to your interface and the input your microphone is on, and set **Monitor** to **In**. Drag Descant onto it. Sing: the meter by **INPUT** should move and the display under it should show the note you're singing. Turn **INPUT** until loud notes just reach the yellow.

3 A track for the notes

Make a MIDI track. Set **MIDI From** to your keyboard and **Monitor** to **In**. Set **MIDI To** to the voice track, and the chooser under it to **Descant**. Play a chord: the keys light up in Descant's **NOTES** strip and the chord is named on the left. Sing, and you hear the vocoder.

4 Add Posy

Put Posy on the same MIDI track. Its chords now go to Descant: press one pad and sing over the whole chord. Load Descant's **Posy Pads** preset to start. Posy's **HOLD** keeps a chord going while your hands are free; so does Descant's own **HOLD**.

5 Set the gate

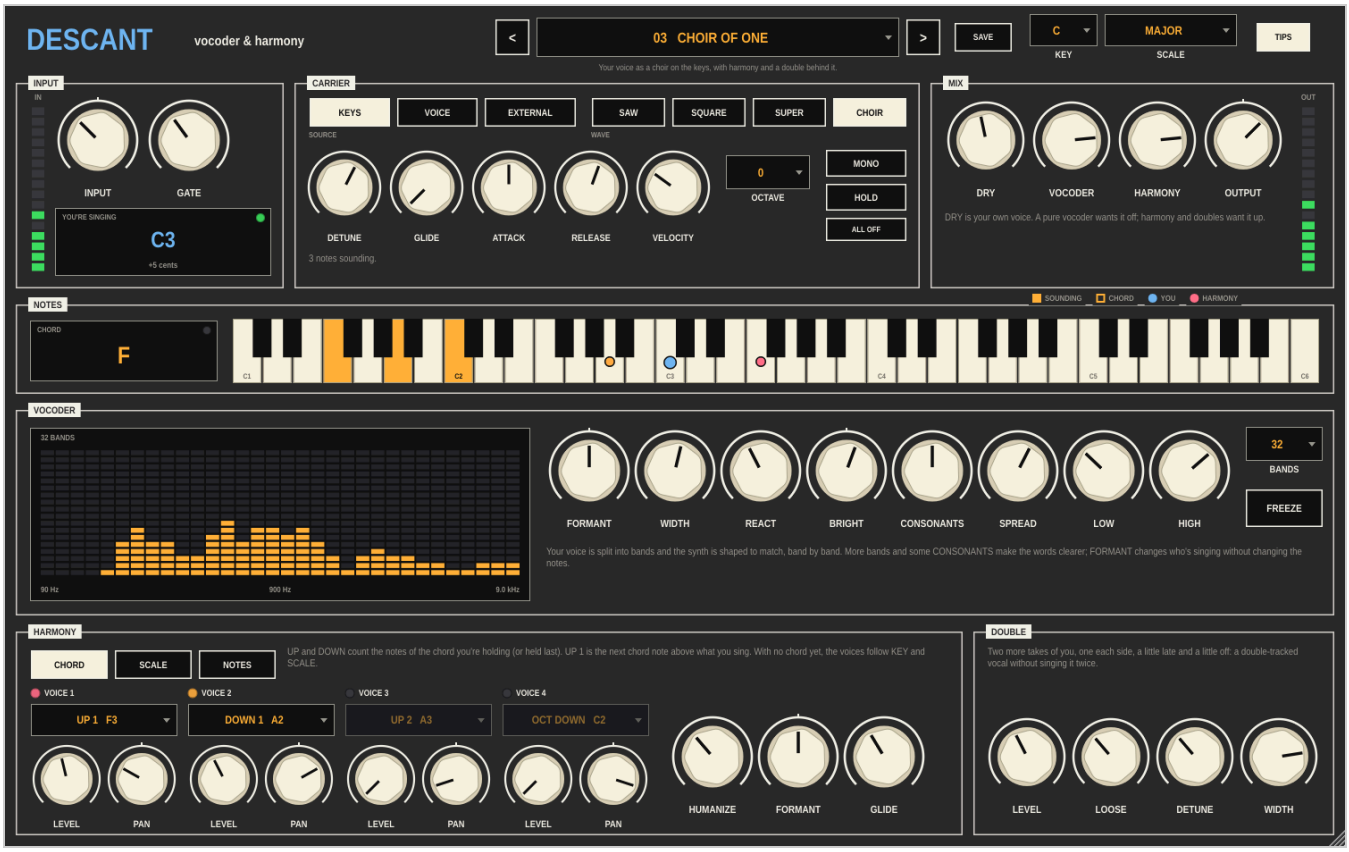
Stop singing and look at the display under INPUT. If it still shows notes, or reads **breath or consonant** while you're quiet, turn **GATE** up until it reads **below the gate**. That stops room noise being vocoded and harmonised.

6 Try the presets

Click the preset name for the list, or step through with **<** and **>**. The line under the name says what each one does. **Choir of One**, **Three Part Harmony**, **Your Voice, Their Chord** and **Robot Voice** show the range.

> **No vocoder?** The vocoder only sounds while notes are held (or with HOLD on). Check the MIDI track's MIDI To points at Descant, and that the light in the CHORD display flickers when you play. The CARRIER section says what Descant is waiting for. Presets that are harmony only have VOCODER turned off in MIX.

The panel



Section	What it does
INPUT	Microphone level, the gate, and the note you're singing (with how many cents off).
CARRIER	The synth your voice shapes. **KEYS** plays the MIDI notes, **VOICE** plays the note you sing snapped to KEY and SCALE, **EXTERNAL** uses another track through the sidechain. SAW, SQUARE, SUPER, CHOIR; MONO, HOLD.
MIX	DRY (your own voice), VOCODER, HARMONY, OUTPUT. A pure vocoder wants DRY off; harmony wants it up.
NOTES	The incoming chord, named, and a keyboard: amber keys sounding, outlined keys the remembered chord, a blue dot for you and a coloured dot for each harmony voice.
VOCODER	BANDS, FORMANT (who's singing, not which note), WIDTH, REACT, BRIGHT, CONSONANTS (clear words), SPREAD, LOW and HIGH, and FREEZE to hold a vowel.
HARMONY	Four voices made from your voice, each with an interval, LEVEL and PAN. HUMANIZE makes them sound like other people; FORMANT makes them bigger or smaller; GLIDE sets how fast they change note.
DOUBLE	Two more takes of you, one each side: LEVEL, LOOSE, DETUNE, WIDTH.

Harmony

The harmony voices are made from your own voice, so they keep your words and your tone. Pick how they choose their notes with the three buttons:

Mode	How the voices pick notes	Use it for
CHORD	**UP 1** is the next note of the chord above what you sing, **UP 2** the one above that; **DOWN** counts down. The chord is the one you're holding, or the last one you held.	Backing vocals that follow the chords from your keyboard or Posy.
SCALE	Fixed steps in KEY and SCALE: a 3rd, 5th or 6th above you, or a 3rd, 4th or 6th below.	Harmony with no keyboard. Set KEY and SCALE to your song.
NOTES	Voice 1 sings the lowest note you hold, voice 2 the next, and so on.	Turning one sung note into a chord: hold a chord and sing anything.

Each voice has a light and a colour ● ● ● ● that match its dot on the keyboard, so you can see where every voice is singing. A voice with its LEVEL all the way down is off.

Ideas to try

- **Backing vocals from Posy.** *Three Part Harmony*: play Posy pads and sing the melody. The two voices land on the chord notes above and below you, and move when the chord does.
- **Harmony without keys.** *Thirds in Key*: set KEY and SCALE, sing, and a voice follows a third above.
- **Your voice as a pad.** *Your Voice, Their Chord*: hold a chord, sing a long "ooh", move the chord.
- **Robot singer.** *Robot Voice*: no keyboard; the vocoder sings the note you sing, snapped to the key.
- **Frozen vowel.** On *Glass Pad*, sing "ah", press FREEZE, stop singing and play: the vowel sings the keys.
- **A synth that talks.** *Sidechain Synth*: open Descant's sidechain (the triangle at the top of the device), pick a synth track, and talk over it.

> **Harmonies wobble or jump?** They follow the pitch they hear. Sing closer to the microphone, raise GATE so the room is shut out, and keep the speakers down. A short GLIDE makes them tighter; a longer one scoops like a real singer. The harmony voices sit about 10–25 ms behind you, like a backing singer; your dry voice and the vocoder have no delay.